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|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONTACT<br>INFORMATION      | 610-F Science and Research 1<br>Houston,<br>TX, USA - 77204<br><br>Mob: +1-3465451497, +91-9871756332,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Website: <a href="http://bhavaytyagi.com">bhavaytyagi.com</a><br>Linkedin: <a href="http://www.linkedin.com/bhavaytyagi">www.linkedin.com/bhavaytyagi</a><br>✉ E-mail: (mail, physics or podcast)<br>[at]bhavaytyagi[dot]com<br>✉ Work E-mail: <a href="mailto:bttyagi@uh.edu">bttyagi@uh.edu</a> |
| EDUCATION                   | <p><b>University of Houston</b>, College of Natural Sciences and Mathematics<br/> <ul style="list-style-type: none"> <li>Research Assistant, PhD Candidate<br/>Advisor: Prof. Eric R. Bittner</li> </ul> </p> <p><b>Durham University</b>, Ustinov College, Durham, UK<br/> <ul style="list-style-type: none"> <li>M.Sc. in <a href="#">Particles, Strings and Cosmology</a>, Result: <b>Merit</b> – 180 credits.<br/>Thesis: <b>An Introduction to AdS/CFT and the Holographic Entanglement Entropy.</b><br/><i>Durham University, UK</i><br/>Supervisor: Prof. Simon F. Ross, Reader: Prof. Nabil Iqbal</li> </ul> </p> <p><b>Amity University Noida</b>, India.</p> | <p>Aug 2022–Present</p> <p>Oct 2018– Sept 2019</p> <p>Aug 2015–May 2018</p>                                                                                                                                                                                                                       |
| MOST RECENT<br>PUBLICATIONS | <ol style="list-style-type: none"> <li>G. Valdivia-Mera, <b>B. Tyagi</b>, E. R. Bittner and P. Hosur. “The Helical SYK Model and Emergent Infrared Integrability” <a href="#">arXiv</a></li> <li>E. R. Bittner and <b>B. Tyagi</b>. “Clockwork quantum: symmetry, noise, and the emergence of quantum order” <a href="#">Philosophical Magazine</a></li> <li><b>B. Tyagi</b>, F. Suzuki, V. A. Chernyak, and N. A. Sinitsyn. “Asymmetry Amplification by a Nonadiabatic Passage through a Critical Point” <a href="#">Physical Review A</a></li> </ol>                                                                                                                 | <p>2026</p> <p>2025</p> <p>2025</p>                                                                                                                                                                                                                                                               |
|                             | The full list can be found on my <a href="#">Google Scholar</a> or <a href="#">arXiv</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                   |
| CURRENT<br>RESEARCH         | <ul style="list-style-type: none"> <li><b>PhD Thesis</b><br/><i>University of Houston, Texas, USA</i><br/>PhD Advisor: Prof. Eric R. Bittner</li> <li><b>Entanglement Dynamics in the Kicked-Field Ising Model</b><br/>Collaborator: Prof. Tomaž Prosen</li> <li><b>Helical SYK Model in 1+1 Dimension</b><br/>Collaborator (Thesis Committee Member): Prof. Pavan Hosur</li> <li><b>Integrable Models of Interacting Majoranas</b><br/>Collaborator (Thesis Committee Member): Dr. Nikolai A. Sinitsyn</li> </ul>                                                                                                                                                     | <p>January 2024–Present</p> <p>December 2025–Present</p> <p>June 2025–June 2026</p> <p>June 2024–Present</p>                                                                                                                                                                                      |
| ACHIEVEMENTS                | <ul style="list-style-type: none"> <li>European Cooperation in Science and Technology Fellowship</li> <li>APS Science Advocacy Champion Award</li> <li>Cullen Fellowship, University of Houston</li> <li>Best Talk, Physics Research Day 2025, University of Houston</li> <li>Graduate Research Fellow at <i>Los Alamos National Lab</i></li> <li>Awarded Distinction for Masters Thesis</li> <li>Undergraduate Research Fellow <i>Harish Chandra Research Institute</i></li> </ul>                                                                                                                                                                                    | <p>2025</p> <p>2025</p> <p>2025</p> <p>2025</p> <p>2024</p> <p>2020</p> <p>2017</p>                                                                                                                                                                                                               |

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| LEADERSHIP &<br>TEACHING<br>EXPERIENCE | • American Physical Society Advocacy Champion                                           | 2025–2026   |
|                                        | • Chair American Physical Society (APS) Chapter at University of Houston                | 2024–2025   |
|                                        | • Physics 2125 (University Level Classical Mechanics)                                   | 2023–2024   |
|                                        | • Physics 1101 (College Level Classical Mechanics)                                      | 2022–2023   |
|                                        | • Physics 2126 (Topics in Modern Physics:<br>Waves, Quantum Mechanics, Nuclear Physics) | Summer 2023 |

#### TALKS/POSTERS

<sup>†</sup>Invited

10. Talk Title: **Lessons from Fundamental Physics<sup>†</sup>** (Public Talk) January 2026  
Sanskriti School  
*New Delhi, India*
9. Talk Title: **On “Noise-Friendly” Quantum Systems** January 2026  
Indian Institute of Technology  
Delhi, India
8. Talk Title: **On “Noise-Friendly” Quantum Systems** September 2025  
Telluride Science and Research Center  
*Telluride, Colorado, USA*
7. Poster Title: **Asymmetry Amplification by a Non-Adiabatic Passage Through a Critical Point** September 2025  
University of Warsaw  
*Warsaw, Poland*
6. Poster Title: **Noise-Induced Synchronisation and Entanglement** March 2025  
APS Global Physics Summit  
*Anaheim, California, USA*
5. Talk Title: **Asymmetry Amplification by a Non-Adiabatic Passage through a Critical Point** March 2025  
APS Global Physics Summit  
*Anaheim, California, USA*
4. Talk Title: **Asymmetry Amplification by a Non-Adiabatic Passage through a Critical Point** Summer 2024  
CNLS Summer Student Talks  
*Center for Non-Linear Studies, Los Alamos National Lab, NM, USA*
3. Talk Title: **Black Hole Information Problem & Recent Developments<sup>†</sup>** September 2021  
**Quantum Photonics Physics Forum<sup>†</sup>** (Public Talk)  
*Online*
2. Talk Title: **On Quantum Entanglement and the Interpretation of Quantum Mechanics.<sup>†</sup>** (Public Talk) 2020  
Christ University Physics Club  
*Christ University Bangalore, India*
1. Talk Title: **On Relativity and Gravitation: Applications to Modern Physics.** (Public Talk) 2018  
Beyond Portals Lecture Series  
*Beyond Portals HQ, New Delhi, India*

#### EXTRA-CURRICULAR ACTIVITIES

- [The Knowmads Podcast](#)
- [Music Channel](#)
- Co-founder of ‘*Beyond Portals*’. An Organization to Promote, Popularize and Support research in fundamental sciences.
- Science Advocacy and Policy Making